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# Online Examination Report

|                |                       |
|----------------|-----------------------|
| Date:          | 21.07.2014            |
| Attempts:      | 1                     |
| Examinee:      | Kanghuli, Joel        |
| Date of birth: | 9/9/1990              |
| Subject:       | 040-Human Performance |
| Result:        | 37,5 %                |
| Time used:     | 00:00:13 (HH:MM:SS)   |

## Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

## Attachment 1: List of result by question

| CQB Number | Internal Number | Syllabus reference | Chapter                                          | Points | Max. Points |
|------------|-----------------|--------------------|--------------------------------------------------|--------|-------------|
| 1          | 175             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 2          | 162             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 3          | 199             | 40.                | HUMAN PERFORMANCE                                | 1,00   | 1,00        |
| 4          | 184             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 5          | 197             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 6          | 193             | 40.                | HUMAN PERFORMANCE                                | 1,00   | 1,00        |
| 7          | 168             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 8          | 181             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 9          | 196             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 10         | 103             | 40.2               | Basic aviation physiology and health maintenance | 0,00   | 1,00        |
| 11         | 16              | 40.2.1.1           | The atmosphere                                   | 1,00   | 1,00        |
| 12         | 45              | 40.2.1.2           | Respiratory and circulatory systems              | 0,00   | 1,00        |
| 13         | 12              | 40.2.1.2           | Respiratory and circulatory systems              | 1,00   | 1,00        |
| 14         | 99              | 40.2.1.2           | Respiratory and circulatory systems              | 1,00   | 1,00        |
| 15         | 52              | 40.2.1.2           | Respiratory and circulatory systems              | 1,00   | 1,00        |
| 16         | 63              | 40.2.1.2           | Respiratory and circulatory systems              | 0,00   | 1,00        |
| 17         | 10              | 40.2.1.2           | Respiratory and circulatory systems              | 1,00   | 1,00        |
| 18         | 33              | 40.2.1.2           | Respiratory and circulatory systems              | 0,00   | 1,00        |
| 19         | 24              | 40.2.1.2           | Respiratory and circulatory systems              | 0,00   | 1,00        |
| 20         | 36              | 40.2.2.2           | Vision                                           | 1,00   | 1,00        |
| 21         | 42              | 40.2.2.2           | Vision                                           | 0,00   | 1,00        |
| 22         | 96              | 40.2.2.3           | Hearing                                          | 0,00   | 1,00        |
| 23         | 58              | 40.2.2.5           | Integration of sensory inputs                    | 0,00   | 1,00        |
| 24         | 32              | 40.2.3.3           | Problem areas for pilots                         | 0,00   | 1,00        |
| 25         | 31              | 40.2.3.4           | Intoxication                                     | 1,00   | 1,00        |
| 26         | 87              | 40.2.3.4           | Intoxication                                     | 0,00   | 1,00        |
| 27         | 17              | 40.2.3.4           | Intoxication                                     | 1,00   | 1,00        |
| 28         | 121             | 40.3               | BASIC AVIATION PSYCHOLOGY                        | 1,00   | 1,00        |

| CQB Number       | Internal Number | Syllabus reference | Chapter                             | Points       | Max. Points  |
|------------------|-----------------|--------------------|-------------------------------------|--------------|--------------|
| 29               | 2               | 40.3               | BASIC AVIATION PSYCHOLOGY           | 1,00         | 1,00         |
| 30               | 14              | 40.3               | BASIC AVIATION PSYCHOLOGY           | 1,00         | 1,00         |
| 31               | 75              | 40.3.1.4           | Response selection                  | 0,00         | 1,00         |
| 32               | 109             | 40.3.3             | Decision making                     | 0,00         | 1,00         |
| 33               | 111             | 40.3.3             | Decision making                     | 1,00         | 1,00         |
| 34               | 26              | 40.3.4.4           | Communication                       | 0,00         | 1,00         |
| 35               | 119             | 40.3.5             | Human behaviour                     | 0,00         | 1,00         |
| 36               | 114             | 40.3.5             | Human behaviour                     | 1,00         | 1,00         |
| 37               | 120             | 40.3.5             | Human behaviour                     | 0,00         | 1,00         |
| 38               | 92              | 40.3.5.1           | Personality, attitude and behaviour | 0,00         | 1,00         |
| 39               | 25              | 40.3.5.1           | Personality, attitude and behaviour | 0,00         | 1,00         |
| 40               | 48              | 40.3.6.2           | Stress                              | 0,00         | 1,00         |
| <b>Total:38%</b> |                 |                    |                                     | <b>15,00</b> | <b>40,00</b> |

## Attachment 2: List of result by topic

**Licence**

**Date: 18.07.2014**

**Examinee: Kanghuli, Joel**

**Location: Uganda Civil Aviation Authority**

**Your Result: 15,00 from 40,00 Points (37,50 %)**

| Topics                                                 | Ammount of Questions | Points achieved | Maximum Points | Percent      |
|--------------------------------------------------------|----------------------|-----------------|----------------|--------------|
| 40. HUMAN PERFORMANCE                                  | 40                   | 15,00           | 40,00          | 37,50        |
| 40.2. Basic aviation physiology and health maintenance | 18                   | 8,00            | 18,00          | 44,44        |
| 40.3. BASIC AVIATION PSYCHOLOGY                        | 13                   | 5,00            | 13,00          | 38,46        |
| <b>Total</b>                                           | <b>40</b>            | <b>15,00</b>    | <b>40,00</b>   | <b>37,50</b> |

1

**Examples of classic behavioral traps that experienced pilots may fall into are to**

- ☒ **promote situational awareness and then necessary changes in behavior.**
- ☐ **assume additional responsibilities and assert PIC authority.**
- ☐ **complete a flight as planned, please passengers, meet schedules, and 'get the job done.'**

2

**What physical change would most likely occur to occupants of an unpressurized aircraft flying above 15,000 feet without supplemental oxygen?**

- ☒ **The pressure in the middle ear becomes less than the atmospheric pressure in the cabin.**
- ☐ **Gases trapped in the body contract and prevent nitrogen from escaping the bloodstream.**
- ☐ **A blue coloration of the lips and fingernails develop along with tunnel vision.**

3

**Why is hypoxia particularly dangerous during flights with one pilot?**

- ☒ Symptoms of hypoxia may be difficult to recognize before the pilot's reactions are affected.
- ☐ The pilot may not be able to control the aircraft even if using oxygen.
- ☐ Night vision may be so impaired that the pilot cannot see other aircraft.

4

**What is the antidote for a pilot with a 'macho' attitude?**

☐ I'm not helpless. I can make a difference.

☐ Taking chances is foolish.

☒ Follow the rules. They are usually right.



5

**An abrupt change from climb to straight-and-level flight can create the illusion of**

- ☒ **tumbling backwards.**
- ☐ **a descent with the wings level.**
- ☒ **a noseup attitude.**

6

Due to visual illusion, when landing on a narrower-than-usual runway, the aircraft will appear to be

- ☐ lower than actual, leading to a higher-than-normal approach.
- ☒ higher than actual, leading to a lower-than-normal approach.
- ☐ higher than actual, leading to a higher-than-normal approach.

7

**What suggestion could you make to people who are experiencing motion sickness?**

- ☒ **Have the students lower their head, shut their eyes, and take deep breaths.**
- ☐ **Avoid unnecessary head movement and focus sight on a point outside the aircraft.**
- ☐ **Recommend taking medication to prevent motion sickness.**

8

**In the aeronautical decision making (ADM) process, what is the first step in neutralizing a hazardous attitude?**

- ☒ Recognizing hazardous thoughts.
- ☐ Making a rational judgement.
- ☒ Recognizing the invulnerability of the situation.

9

**A sloping cloud formation, an obscured horizon, and a dark scene spread with ground lights and stars can create an illusion known as**

☐ **false horizons.**

☒ **autokinesis.**

☐ **elevator illusions.**

10

Which of these is a common symptom of hyperventilation?

☒ Decreased breathing rate.

☐ A sense of well-being.

☐ Drowsiness.

11

Which gas below 70,000 ft., above ground gas makes up the major part of the atmosphere ?

☒ Nitrogen

☐ Carbon dioxide

☐ Ozone

☐ Oxygen

12

The total gas volume of the lung is the sum of:

1. tidal volume
2. inspiratory reserve volume
3. expiratory reserve volume
4. residual volume

Which of the following lists the correct combination?

- ☒ 2 and 3
- ☐ 1 and 2
- ☐ 1, 2, 3 and 4
- ☐ 1, 2 and 3



13

Hypoxia is caused by

- ☐ reduced partial pressure of nitrogen in the lung
- ☐ a higher affinity of the red blood cells (haemoglobin) to oxygen
- ☒ reduced partial oxygen pressure in the lung
- ☐ an increased number of red blood cells

14

**Hypoxia can be caused by**

- ☐ increasing oxygen partial pressure used for the exchange of gases
- ☒ decreased ability of the haemoglobin to transport oxygen
- ☐ too much carbon dioxide in the blood
- ☐ lack of nitrogen in ambient air

15

What can be said concerning the following two statements?

1. Euphoria can be a symptom of hypoxia.
2. Someone in an euphoric condition is more prone to error.

☐ 1 is correct, 2 is false

☐ 1 and 2 are both false

☐ 1 is false, 2 is correct

☒ 1 and 2 are both correct

16

**When flying above 10.000 feet hypoxia arises because:**

- ☐ composition of the blood changes
- ☒ percentage of oxygen is lower than at sea level
- ☐ partial oxygen pressure is below the critical value of 55mm/Hg
- ☐ composition of the air is different from sea level

17

The rate and depth of breathing is primarily regulated by the concentration of:

- ☐ water vapour in the alveoli
- ☐ nitrogen in the air
- ☒ carbon dioxide in the blood
- ☐ oxygen in the cells

18

**Hyperventilation is:**

- ☐ an increased heart rate caused by a decreasing blood-pressure
- ☐ an increased heart rate caused by an increasing blood pressure
- ☐ a normal compensatory physiological reaction to a drop in partial oxygen pressure (i.e. when climbing a high mountain)
- ☒ a reduction of partial oxygen pressure in the brain

19

**When faced with sustained cold temperature, how does the body resist this physical stress?**

- ☐ By increasing cardiac frequency.
- ☐ By intense vasoconstriction.
- ☒ By speeding up the metabolic rate in the Autonomic Nervous System.
- ☐ By vasodilatation which permits a greater flow of blood to the periphery.

20

**Presbyopia:**

- ☐ is caused by long-termed exposure to stimuli over 90dB.
- ☐ surgical replacement of the lens the usual treatment and is compatible with flying.
- ☒ is common over the age of 50.
- ☐ is partial visual loss due to pressure changes in the eye.



21

**When flying through a thunderstorm with lightning you can protect yourself from flashblindness by:**

- 1 - turning up the intensity of cockpit lights**
- 2 - looking inside the cockpit**
- 3 - wearing sunglasses**
- 4 - using blinds or curtains when installed**

☒ **3 and 4 are correct, 1 and 2 are false**

☐ **1, 2, 3 and 4 are correct**

☐ **1 and 2 are correct, 3 and 4 are false**

☐ **1, 2 and 3 are correct, 4 is false**

22

**Noise induced hearing loss:**

- ☐ is a condition resulting in permanent hearing loss of selected frequencies.
- ☒ is also known as presbycusis and is associated with pressure damage to the middle ear.
- ☐ causes Eustachian Tube dysfunction.
- ☐ is not a permanent hearing loss, the nerve cells frequently recover.

23

What would be the effect if, in a tight turn, one bends down to pick up a pencil?

☐ Barotrauma.

☒ Inversion Illusion.

☐ Vertigo.

☐ Coriolis effect.

24

**Flying with a "common cold":**

- ☐ may lead to incapacitation due to severe sinus or ear pain.
- ☐ will cause infection in other crew members if you are flying in a pressurised aircraft.
- ☒ increases the risk of hypertension.
- ☐ is permitted as long as you are on treatment with antibiotics.

25

Alcohol, when taken simultaneously with drugs, may

- ☐ compensate for side effects of drugs
- ☐ increase the rate of drug elimination from the blood
- ☐ lead to undesired effects only during night flights
- ☒ intensify the side effects of the drugs

26

Which statement is correct?

1. Smokers have a greater chance of suffering from coronary heart disease
2. Smoking tobacco will raise the individuals physiological altitude during flight
3. Smokers have a greater chance of contracting lung cancer

☐ 1 and 2 are correct, 3 is false

☐ 2 and 3 are correct, 1 is false

☐ 1,2 and 3 are correct

☒ 1 and 3 are correct, 2 is false

27

**Carbon monoxide (CO) poisoning in flight:**

- ☒ presents an extremely dangerous situation as the blood may not be able carry sufficient amounts of oxygen to vital cells and tissues of the body.
- ☐ is usually harmless because oxygen is more easily attached to haemoglobin than carbon monoxide to a magnitude of 200 times.
- ☐ is a complication when hyperventilating and requires its own special and individual treatment.
- ☐ can be cured by breathing into a plastic bag to retain the carbon monoxide.

28

To help manage cockpit stress, pilots must

- ☒ condition themselves to relax and think rationally when stress appears.
- ☐ avoid situations that will improve their abilities to handle cockpit responsibilities.
- ☐ be aware of life stress situations that are similar to those in flying.



29

What are two types of attention ?

☐ Intuitive and behavioural

☐ Cognitive and intuitive

☒ Selective and divided

☐ Divided and behavioural

30

**Carbon Monoxide is particularly dangerous because:**

1. Its initial symptoms are not alarming
2. It is colourless
3. Its is odourless
4. It is highly toxic
5. Its effects are cumulative

☐ 2, 3, and 4 only

☐ 1, 2, 3 and 5 only

☒ all of the above

☐ 2, 3, 4 and 5 only

31

**Planning:**

- ☐ allows crew members to anticipate potential risky situations and decide on possible responses
- ☒ in the cockpit typically results in plans that are always easy to modify when things are not as anticipated
- ☐ is dangerous in the cockpit, as it interrupts flight crew creativity
- ☐ is unnecessary in the cockpit, as crew members are so highly trained, they will always know what to do in unusual situations

32

on which features does risk management, as part of the Aeronautical Decision Making (ADM) process, rely?

- ☒ The mental process of analyzing all information in a particular situation and making a timely decision on what action to take.
- ☐ Situational awareness, problem recognition, and good judgment.
- ☐ Application of stress management and risk element procedures.

33

Which of the following is considered as a step involved in good Aeronautical Decision Making (ADM) process?

- ☐ making a rational evaluation of the required actions.
- ☒ identifying personal attitudes hazardous to safe flight.
- ☐ developing the 'right stuff' attitude.

34

Which of the following are strategies for resolving conflict?

1. Seeking arbitration
2. Actively listening to other people
3. Recognise the early signs of conflict and address them
4. Becoming aware of cultural influences

☒ 1, 2, 3 only

☐ 1, 2, 4 only

☐ 1, 2, 3 and 4

☐ 2, 3, 4 only

35

**What is the first step in neutralizing a hazardous attitude in the ADM process?**

☒ **Recognition of invulnerability in the situation.**

☐ **Recognition of hazardous thoughts.**

☐ **Dealing with improper judgment.**

36

**Some of these dangerous tendencies or behavior patterns which must be identified and eliminated include:**

- ☐ Deficiencies in instrument skills and knowledge of aircraft systems or limitations.
- ☒ Peer pressure, get-there-itis, loss of positional or situation awareness, and operating without adequate fuel reserves.
- ☐ Performance deficiencies from human factors such as, fatigue, illness or emotional problems.



37

**What should a pilot do when recognizing a thought as hazardous?**

☐ Develop this hazardous thought and follow through with modified action.

☒ Avoid developing this hazardous thought.

☐ Label that thought as hazardous, then correct that thought by stating the corresponding learned antidote.

38

Which of the following elements make up the personality of an individual?

1. Heredity
2. Childhood environment
3. Upbringing
4. Past experience

☐ 1,2,4

☒ 2,3,4

☐ 1,2,3,4

☐ 2,3

39

**Contrary to a person's personality, attitudes:**

- ☐ form part of personality and, as a result, cannot be changed in an adult
- ☐ are essentially driving forces behind changes in personality
- ☒ are non-evolutive adaptation procedures regardless of the result of the actions associated with them
- ☐ Are the product of personal disposition and past experience with reference to an object or a situation

40

**The behavioural effects of stress may include:**

- 1: manifestation of aggressiveness.
- 2: a willingness to improve communication.
- 3: a willingness for group cohesion.
- 4: a tendency to withdrawal.

**The combination of correct statements is:**

☐ 1,2 and 3 are correct

☐ 3 and 4 are correct

☐ 1 and 4 are correct

☒ 2,3 and 4 are correct